Special Ship Automation Systems HW: System Modeling Using Experimental Measurement Data

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The Problem

- Experiment measures
- Operation measures
- Virtual sensors
- Digital twins

Data Analysis

The Problem

Data Analysis

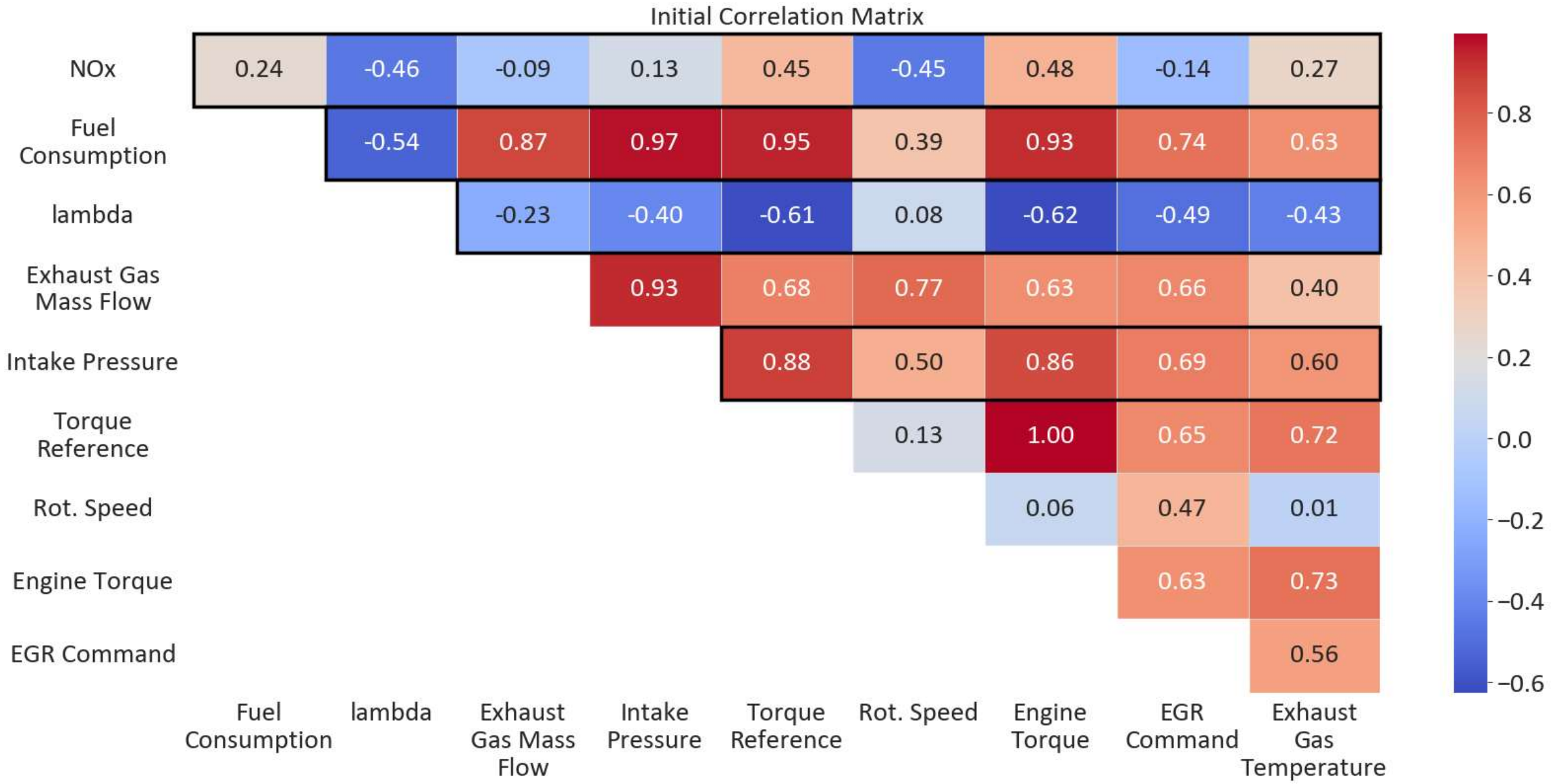
The Neural Network

Evaluation

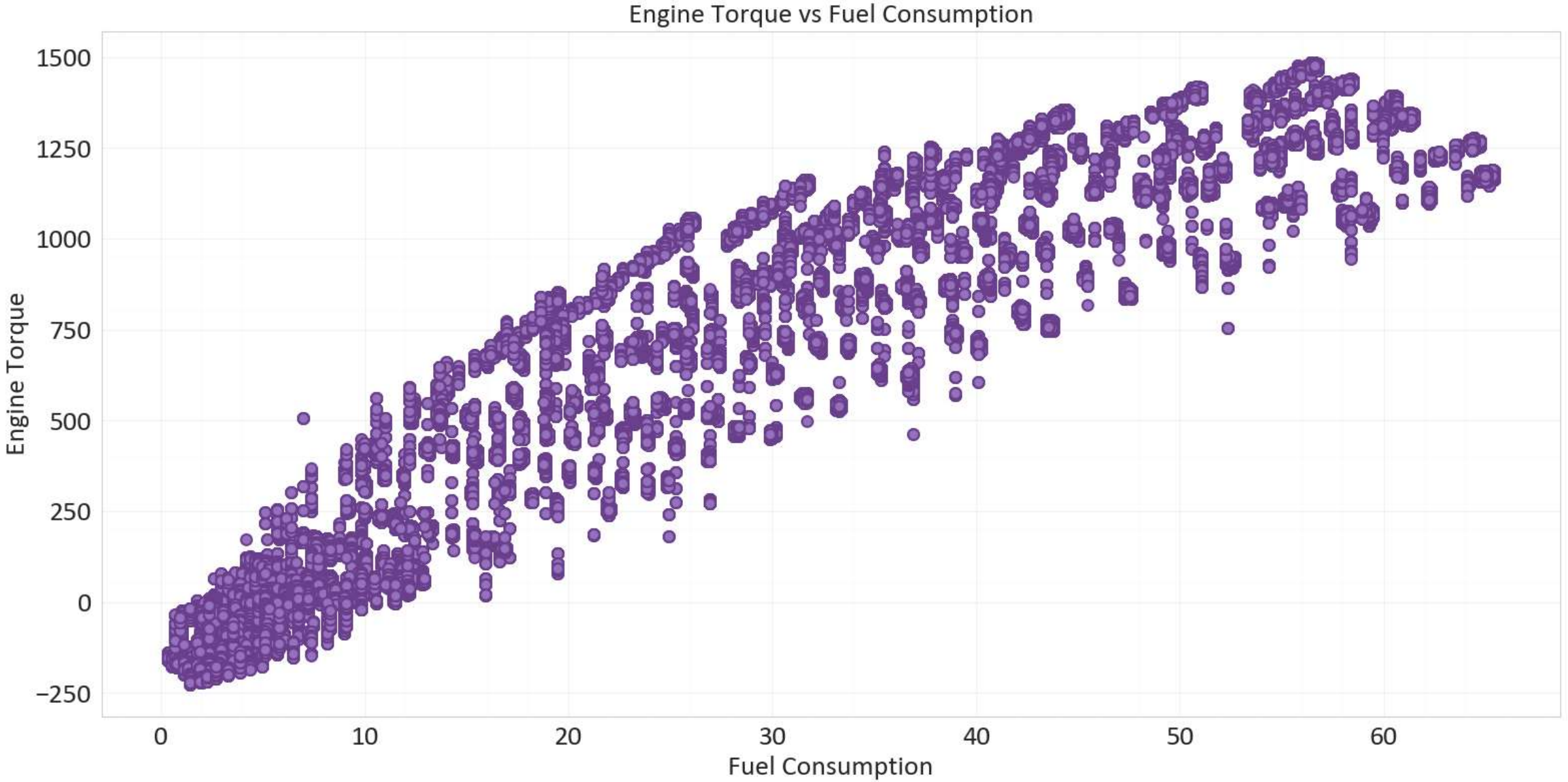
Feature Engineering

Optimization

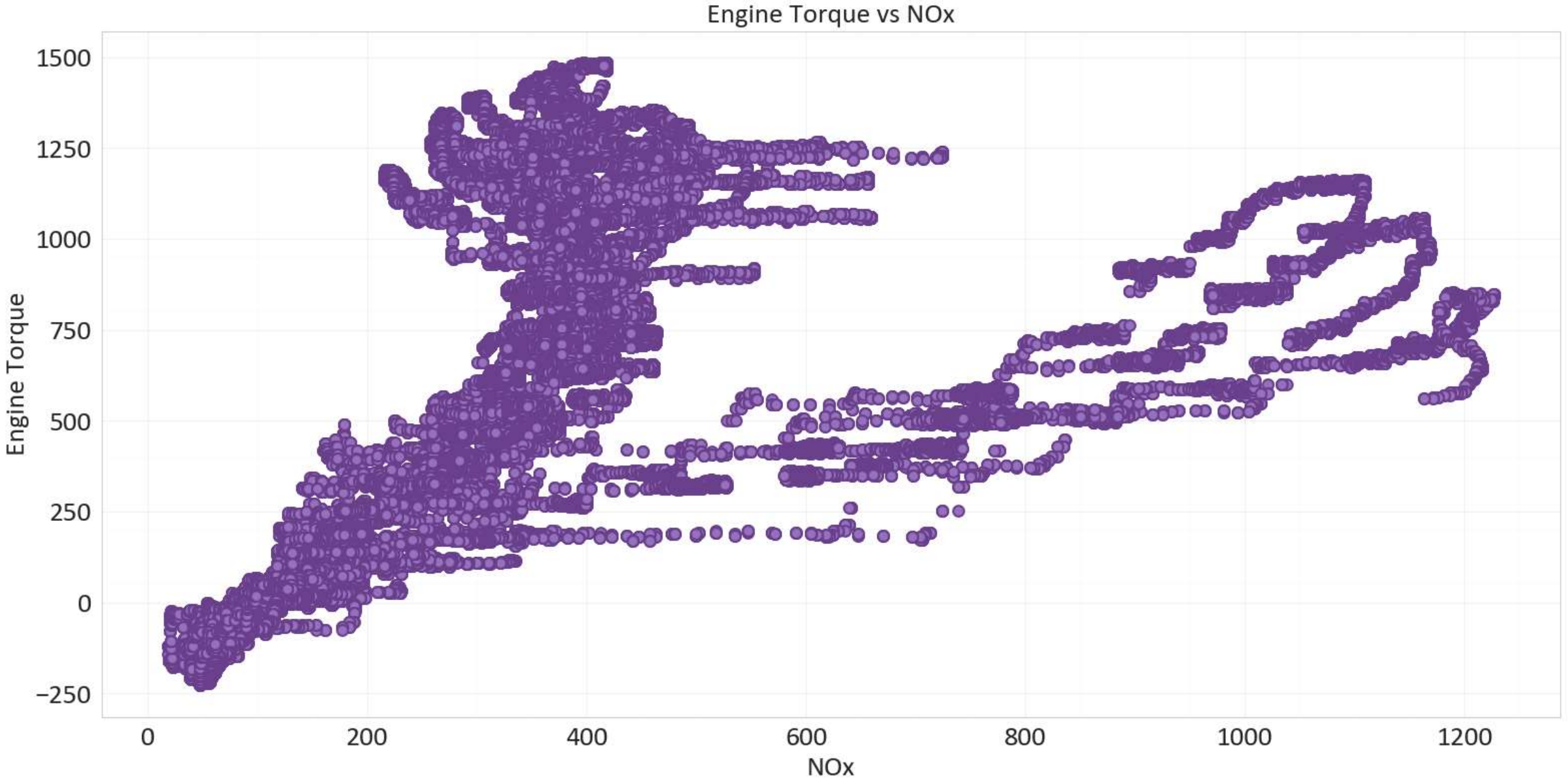
Data Analysis



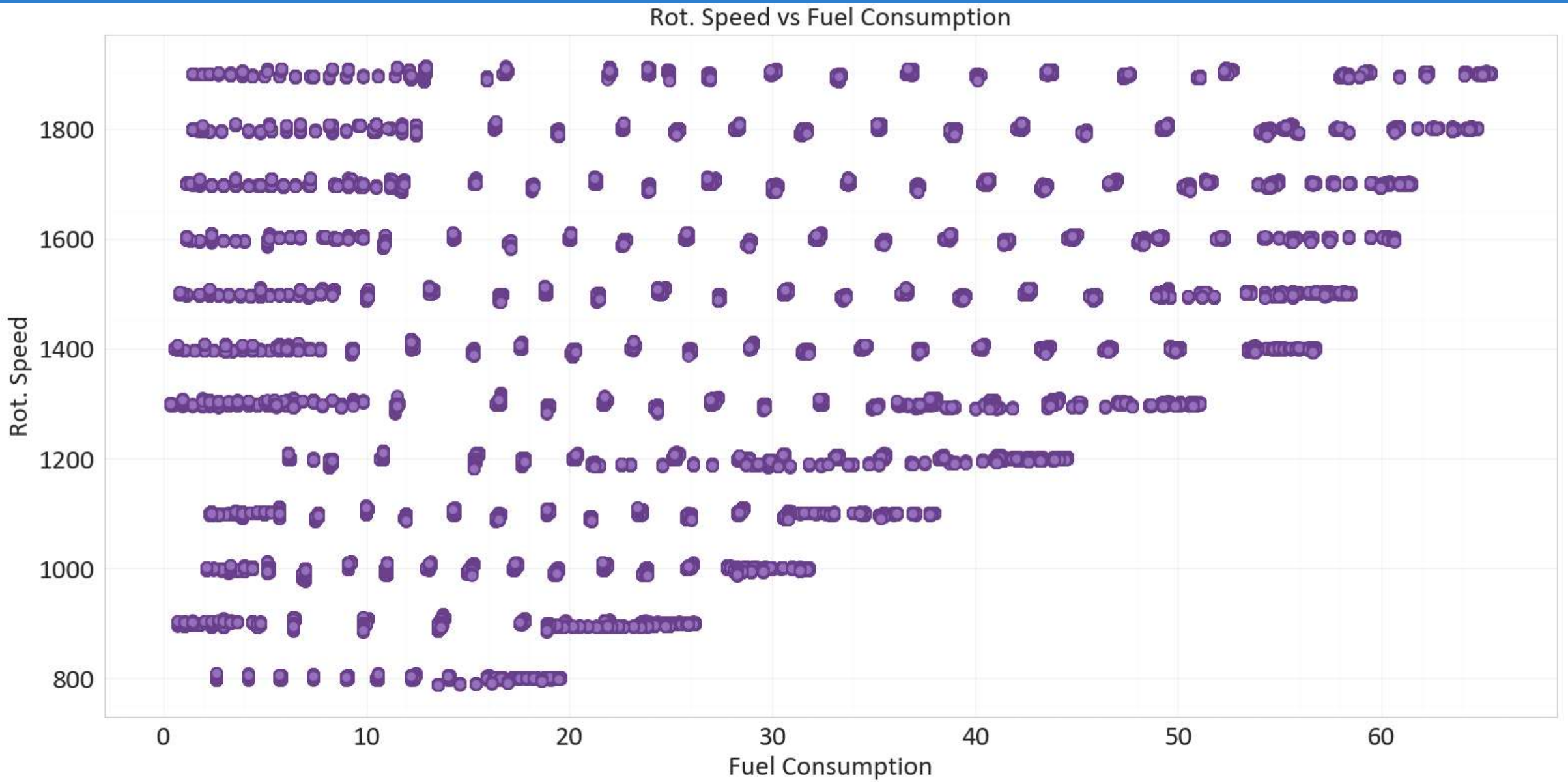
Data Analysis



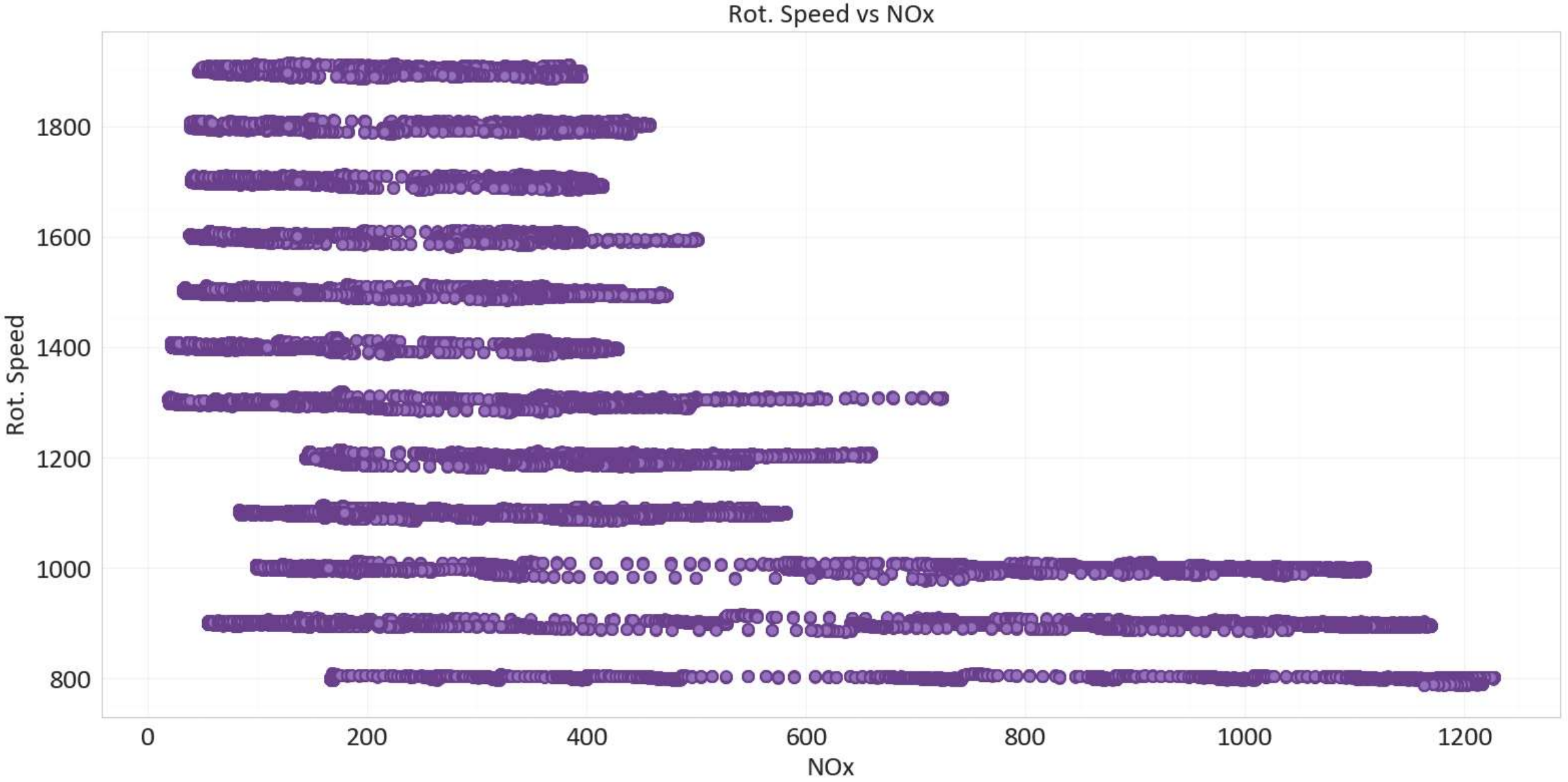
Data Analysis



Data Analysis



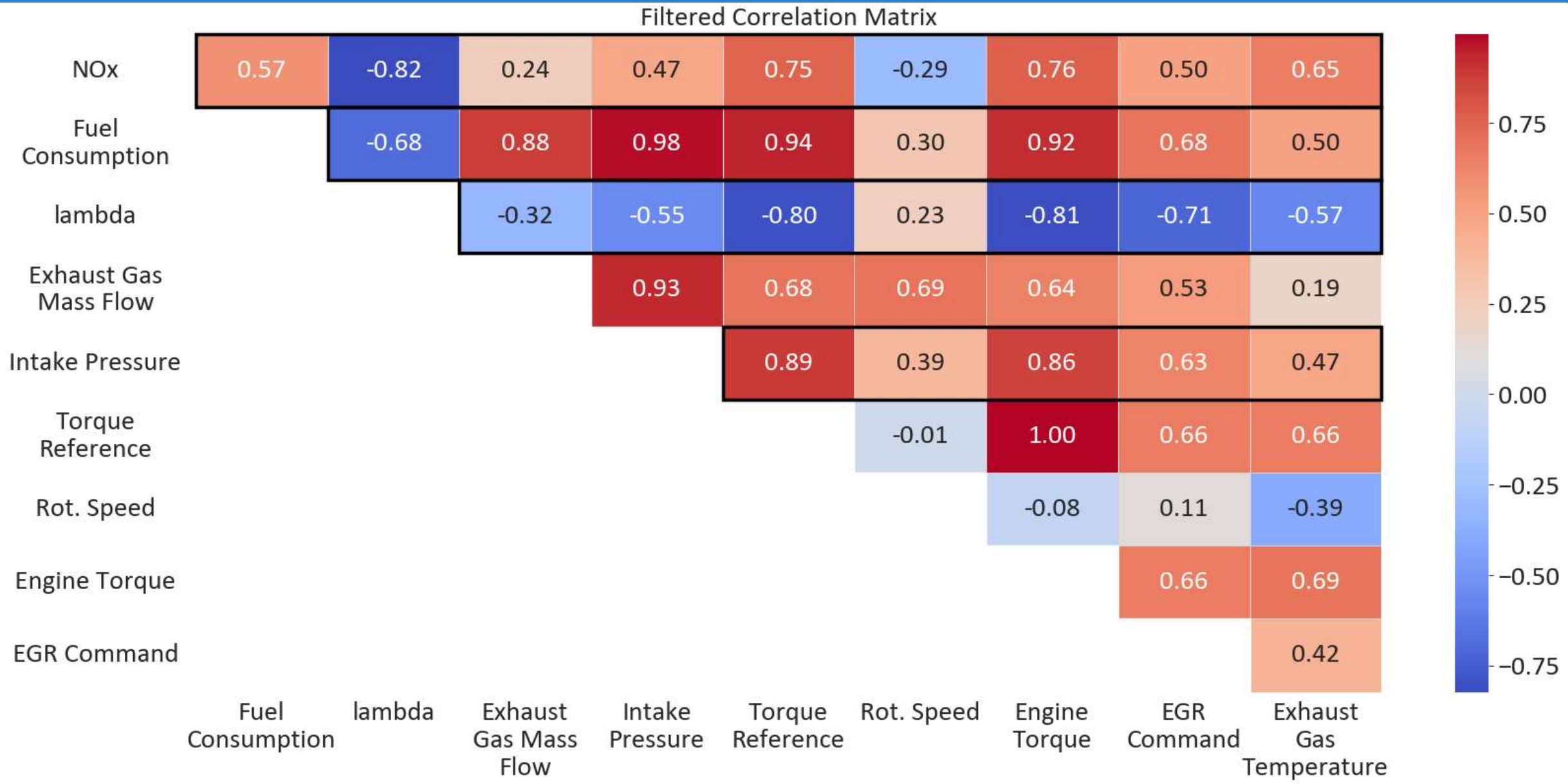
Data Analysis



Data Analysis

- **Data Filtering**
  - Values > 200% mean
  - Duplicates
  - Negatives

Data Analysis



The Neural Network

The Problem

Data Analysis

The Neural Network

Evaluation

Feature Engineering

Optimization

The Neural Network

Exhaust Gas Mass Flow

Rot. Speed

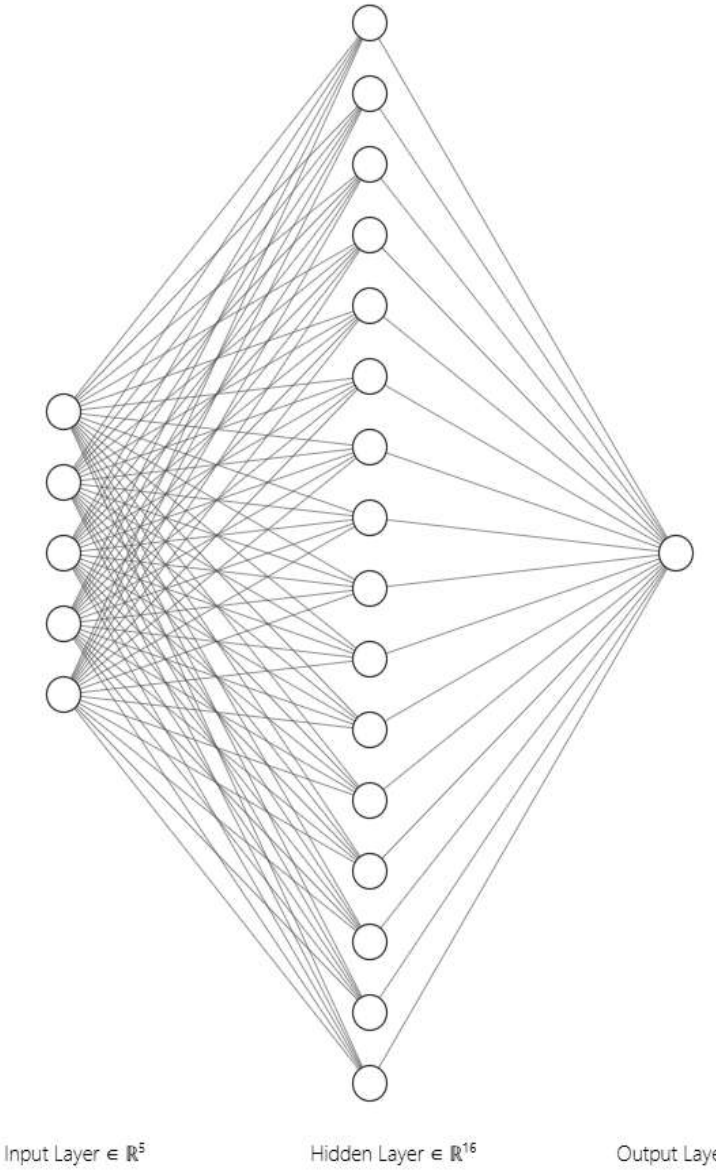
Engine Torque

EGR Command

Exhaust Gas Temperature

} NOx / Fuel Consumption

- 16 Neurons
- ReLU
- MSE Loss
- Discrete RPM groups (1100-1900)
- Input clustering according to the RPM ranges



The Neural Network

- Data split: 70% training, 10% validation and 20% test set
- 500 epochs
- Standard Scaler
- Data conversion to tensors, compatibility with PyTorch
- Adam optimizer - lr=0,01

$$z = \frac{x - \mu}{\sigma}$$

Evaluation

The Problem

Data Analysis

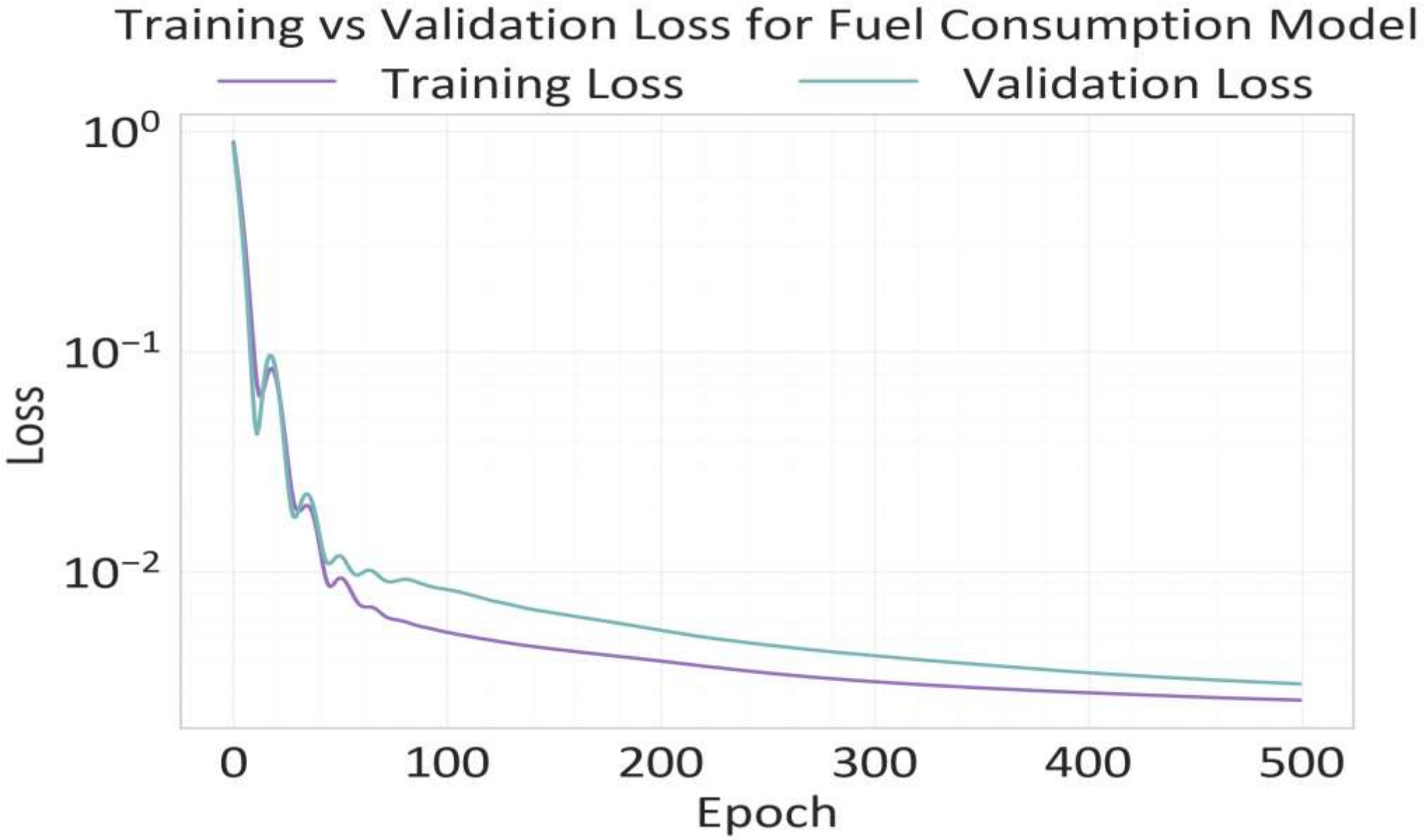
The Neural Network

Evaluation

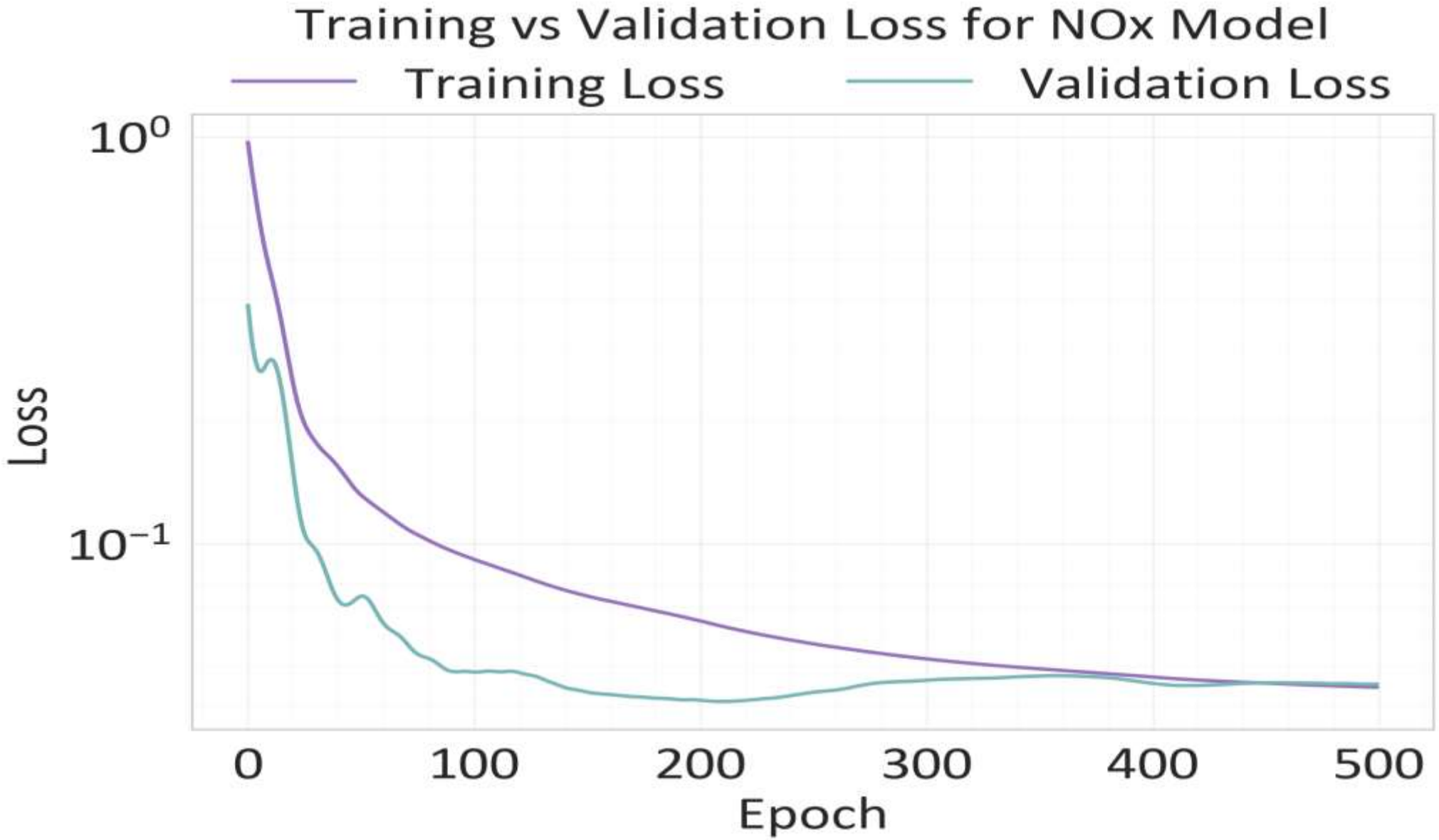
Feature Engineering

Optimization

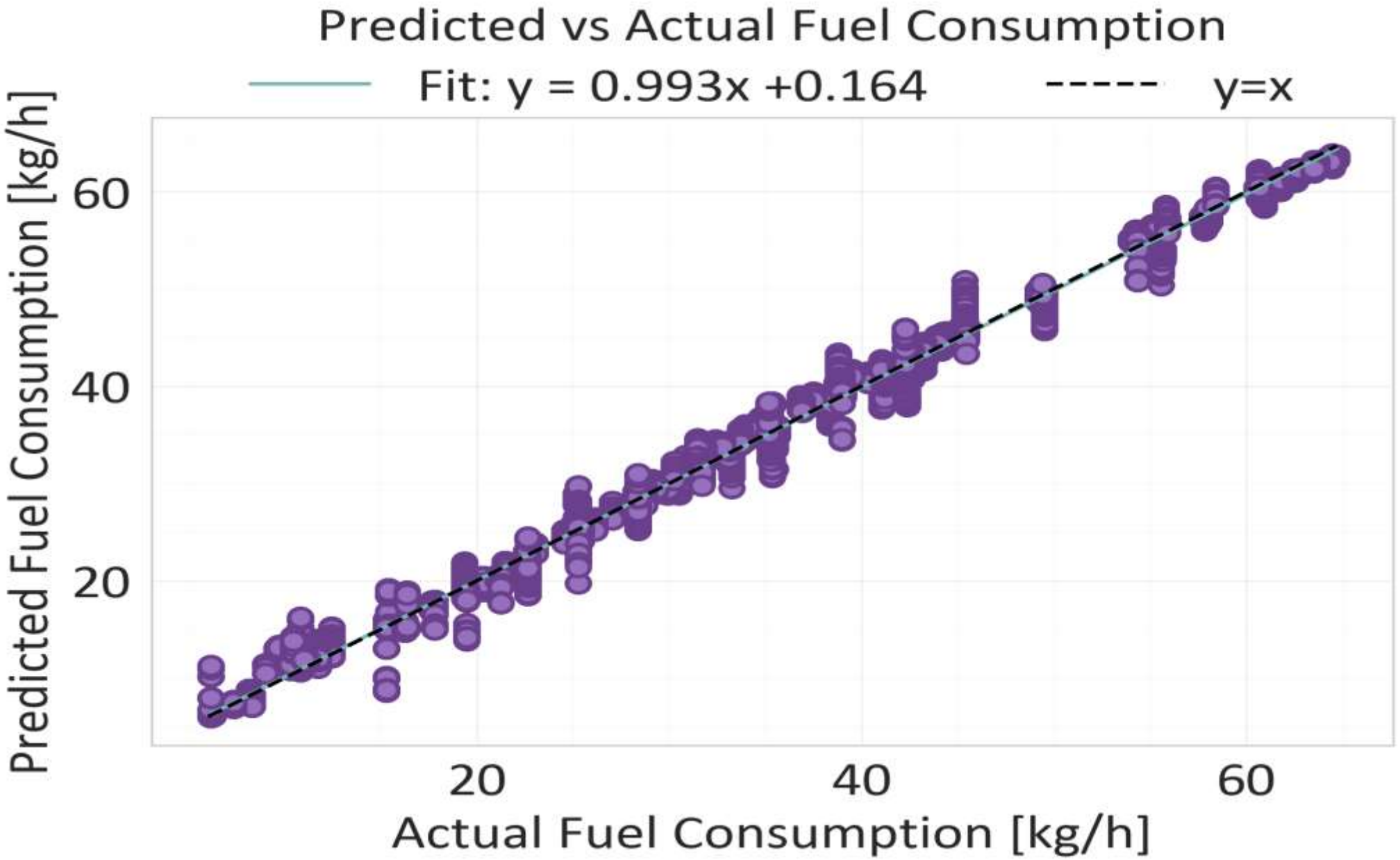
Evaluation


Training vs Validation Loss for Fuel Consumption Model
Training Loss
Validation Loss
10^0
10^{-1}
10^{-2}
Loss
0
100
200
300
400
500
Epoch

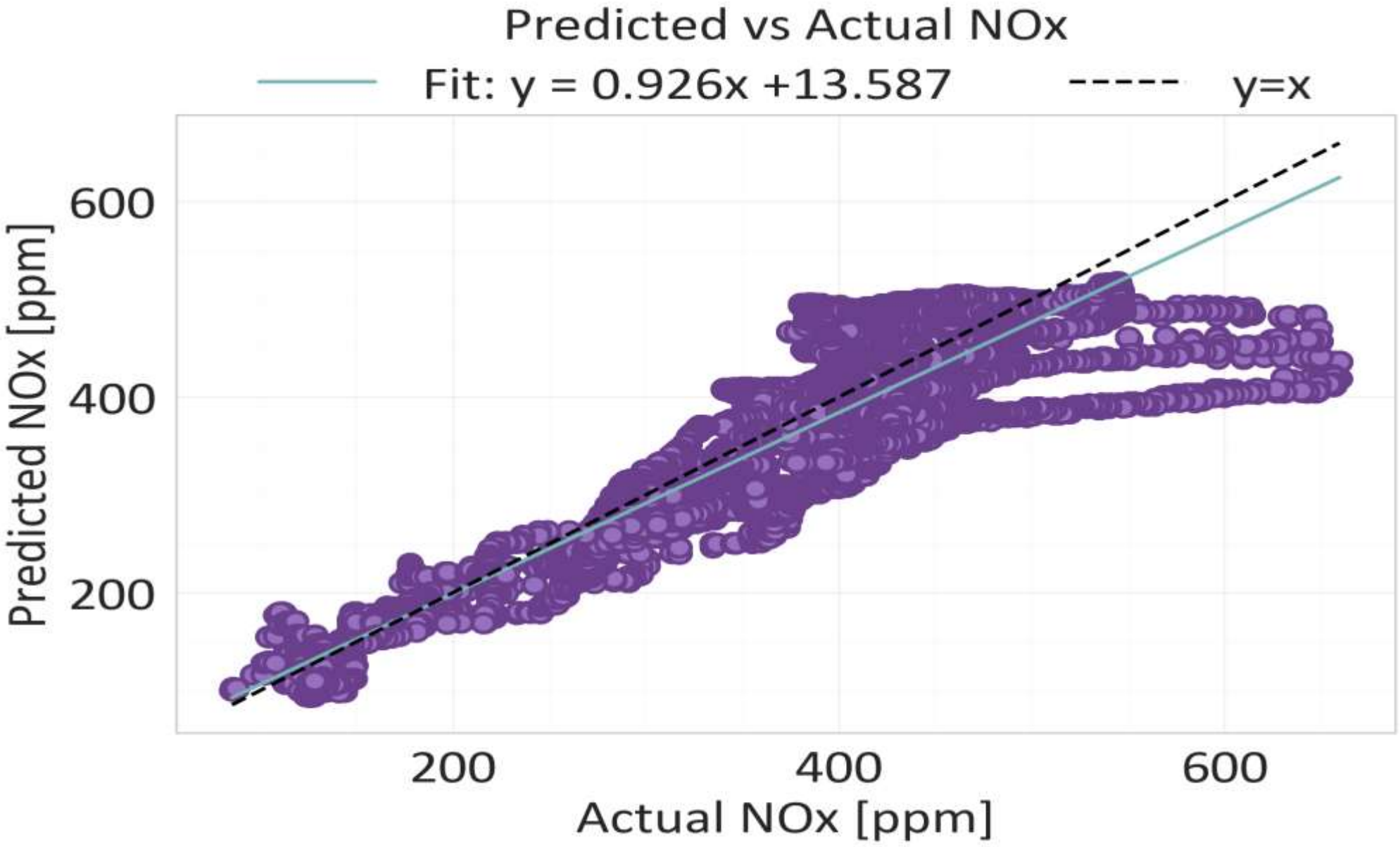
Evaluation


Training vs Validation Loss for NOx Model
Training Loss
Validation Loss
10^0
10^{-1}
Loss
0
100
200
300
400
500
Epoch

Evaluation


Predicted vs Actual Fuel Consumption
Fit: y = 0.993x +0.164
y=x
Predicted Fuel Consumption [kg/h]
60
40
20
20
40
60
Actual Fuel Consumption [kg/h]

Evaluation


Predicted vs Actual NOx
Fit: y = 0.926x +13.587
y=x
Predicted NOx [ppm]
600
400
200
200
400
600
Actual NOx [ppm]

Evaluation

- **Simple model evaluation**

Model	RMSE scaled	RMSE real	Relative RMSE	R^2	L^∞ scaled
Fuel Consumption	0.048	0.742 kg/h	2.12%	0.997	0.431
NOx	0.401	41.19 ppm	10.98%	0.846	2.384

Feature Engineering

The Problem

Data Analysis

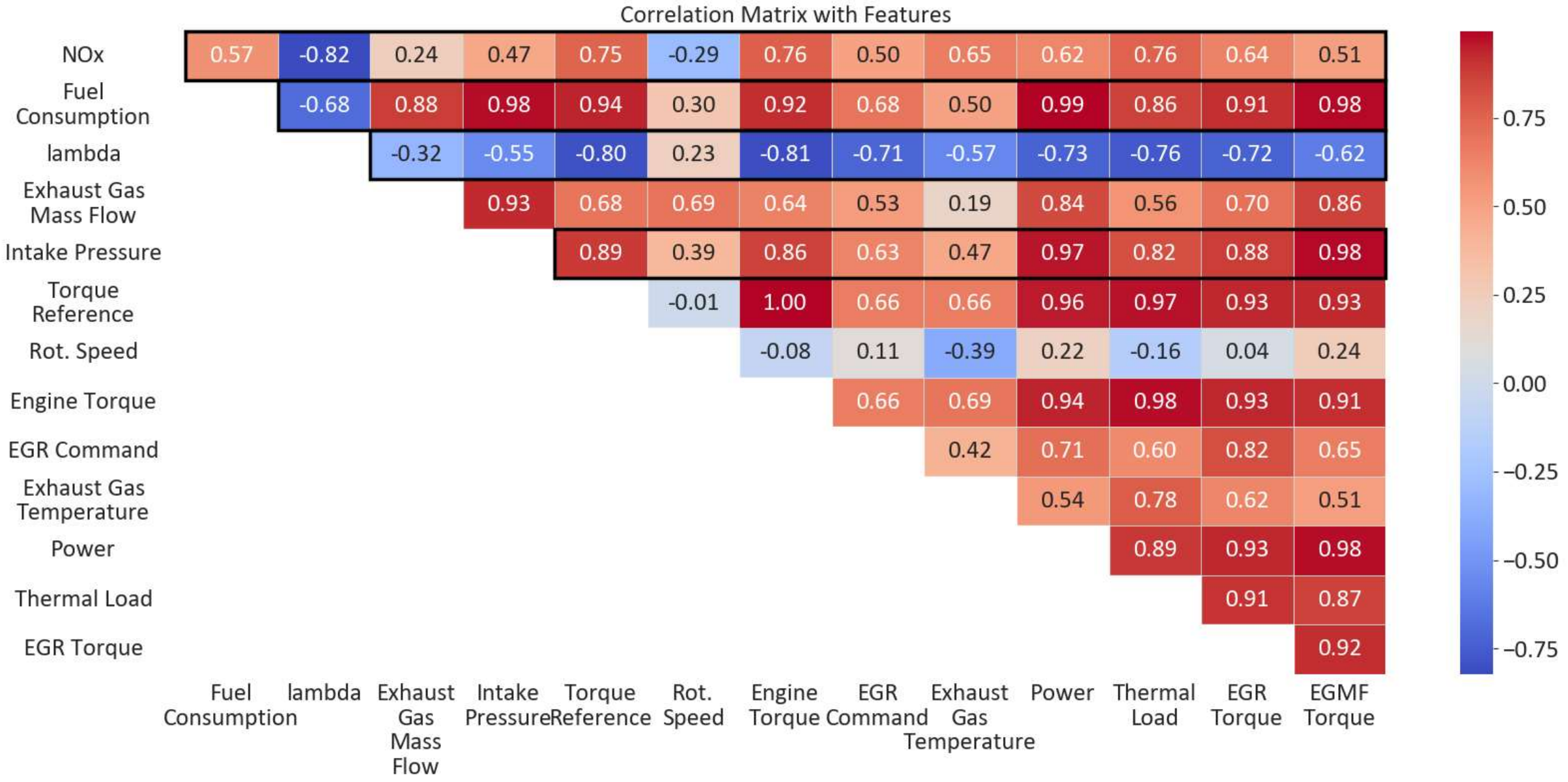
The Neural Network

Evaluation

Feature Engineering

Optimization

Feature Engineering



Feature Engineering

- **Feature engineered model evaluation**

Model	RMSE scaled	RMSE real	Relative RMSE	R^2	L^∞ scaled
Fuel Consumption	0.045	0.690 kg/h	1.97%	0.997	0.457
NOx	0.416	42.75 ppm	11.40%	0.834	2.516

Optimization

The Problem

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Optimization

- WandB grid sweeps
- Test RMSE, R^2 - $L\infty$

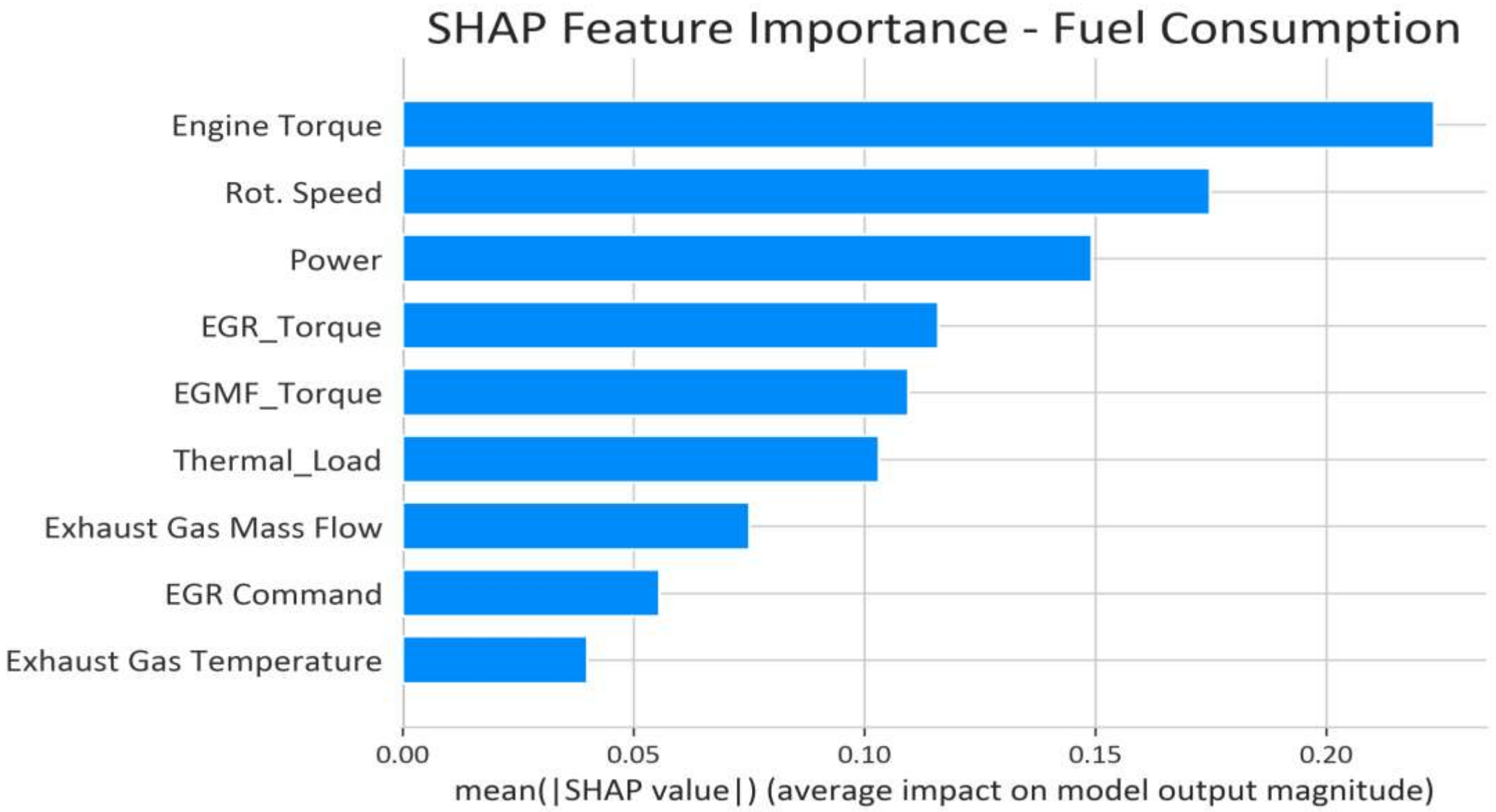
Parameter	Tested values
Simple model neurons	[16], [20]
Complex model neurons	[32], [32,32], [64,64]
Learning rate	0.001, 0.01
Simple epochs	500, 1000
Complex epochs	200, 500

Optimization

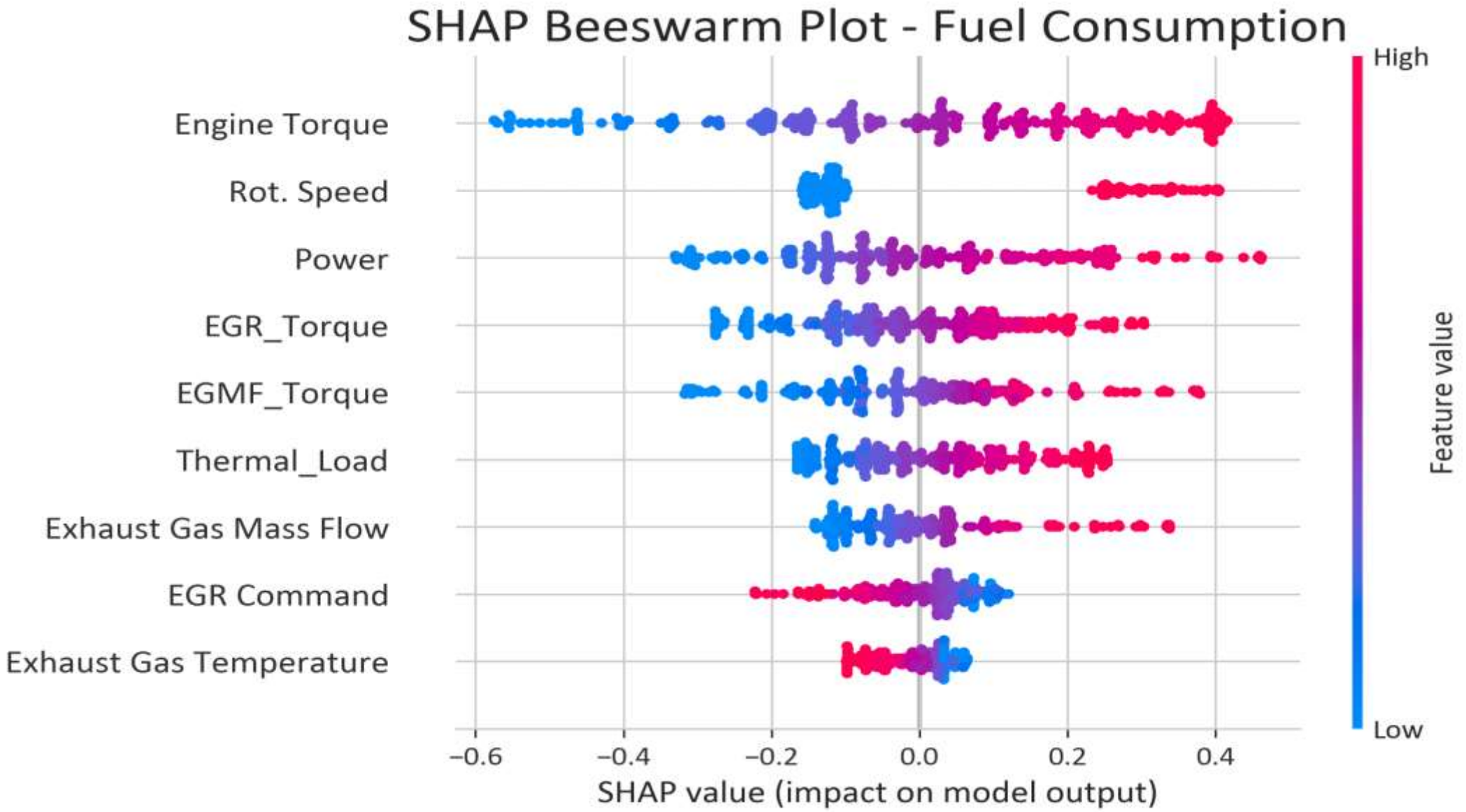
- **Best Models**

Fuel Consumption	NOx
[64, 64]	[64, 64]
0,01	0,001
200	500

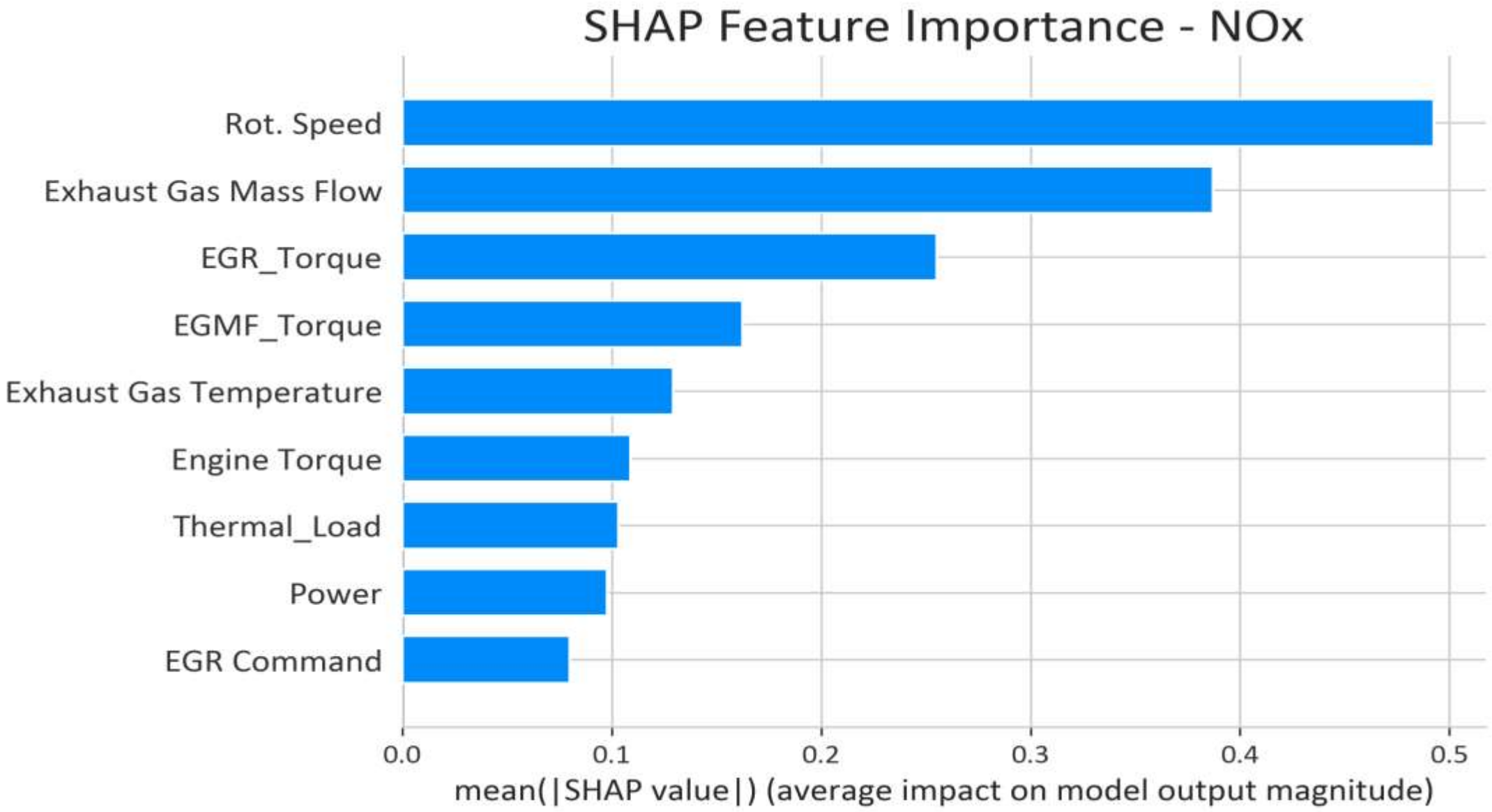
Optimization



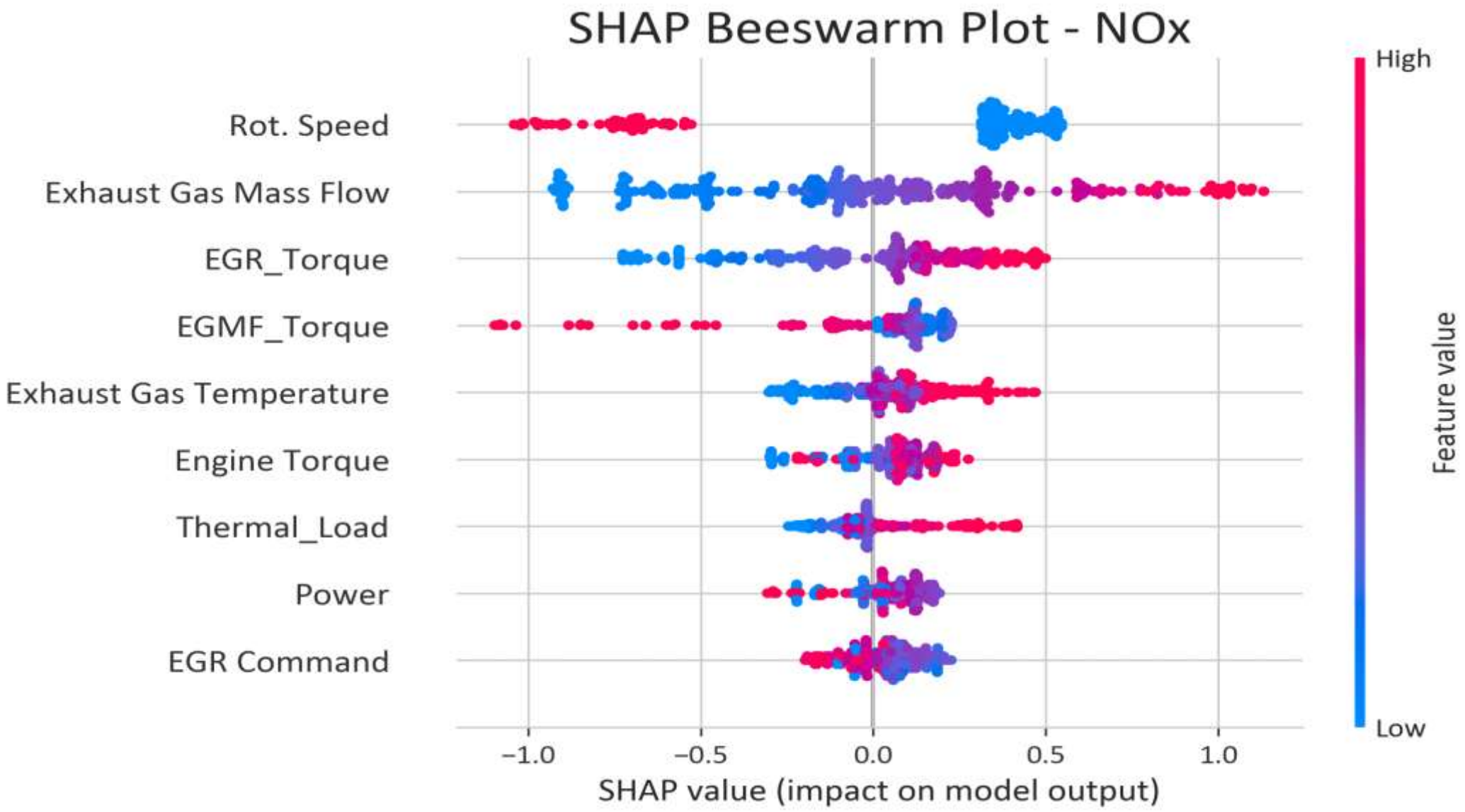
Optimization



Optimization



Optimization



Optimization

- **Complex model evaluation**

Model	RMSE scaled	RMSE real	Relative RMSE	R^2	L^∞ scaled
Fuel Consumption	0.043	0.660 kg/h	1.89%	0.998	0.451
NOx	0.378	38.90 ppm	10.37%	0.863	2.312